

Raise the turnover, hold the length.

Your stroke length is already strong — about **1.79 m per stroke**. Your speed is capped by **turnover**: you cruise near **19.6 strokes/min**, and your rate sags in the back of long swims. This block lifts your stroke rate toward **22–24/min** while protecting the length you've built, pairing three pool sessions a week with two short dryland sessions that raise your power ceiling and keep your shoulders durable. Everything is anchored to one test you run in Week 1.

01 Where you stand

STROKE RATE

19.6

→ 22–24 spm

DIST / STROKE

1.79m

hold ≥ 1.79

PURE PACE / 100M

2:50

→ ~2:21–2:32

SWOLF

~56

→ falls as rate rises

Speed in the water is simply **strokes/min** × **distance/stroke**. You've maxed the second term and let the first drift down. Restoring turnover to your old level while keeping today's stroke length is, by your own numbers, worth roughly 20–30 seconds per 100m.

02 Two tests: Week 1 and Week 8

Critical Swim Speed (your pace anchor)

Warm up well. Swim a **400m** time trial all-out; rest 5+ min; swim a **200m** all-out.

CSS pace /100m = (400m time – 200m time) ÷ 2

Every "@ CSS" below means this pace. Re-test in Week 8 — it should drop. Your best-effort history suggests a starting CSS near **2:21–2:25**; trust the test over the estimate.

Stroke-count baseline (your length guardrail)

On an easy 100m, count strokes per length (you're around **14** for 25m). Note it.

The golden rule of this whole block: when you add tempo, your strokes-per-length must not climb more than a stroke or two. Faster rate at the same count = faster swimming. Faster rate with a ballooning count = thrashing. Watch this number every tempo set.

03 How a week fits together

DAY	SESSION	WHY HERE
Mon	Pool · Threshold/Tempo	Keystone. Freshest day.
Tue	Dryland A · Power & turnover	Explosive; 4 days clear of the speed swim.
Wed	Rest / easy mobility	Absorb the hard start.
Thu	Pool · Technique & aerobic	Low intensity; rebuild length.
Fri	Dryland B · Strength-end & resilience	Lower CNS load; safe before speed.
Sat	Pool · Speed & race-pace	Top-end with legs/arms recovered.
Sun	Full rest	Adaptation happens now.

Template, not law — shift days to your life, but keep two rules: don't put Dryland A (explosive) the day before the speed swim, and keep one easy/rest day before the threshold swim.

Tempo Trainer settings

Mode 1 (seconds per stroke). Set it one notch faster than your natural rhythm and climb gradually. Hold your stroke count.

WEEKS	BEEP (S)	≈ SPM
1–2	2.95	~20.3
3–4	2.85	~21.1
5–6	2.75	~21.8
7	2.65	~22.6
8	free — feel your new rhythm	

One beep = one arm entry. Calibrate to your device; the spm column is the goal.

04 The 8 weeks

Pool mains shown; every session adds the warm-up, drills and cool-down from the library below. "@ CSS" = your tested threshold pace; "+8" = 8s/100m slower, "-5" = 5s faster.

WK	POOL 1 · THRESHOLD	POOL 2 · TECHNIQUE	POOL 3 · SPEED	TEMPO	DRYLAND
1 FOUND.	TEST + SET CSS test (400+200 TT), then 6×100 @ CSS, 20s rest	8×50 drill (count strokes) · 6×100 @ CSS+8, lowest smooth count	12×25 fast, full rest · 4×100 @ CSS	2.95s ~20 spm	A & B at base loads — learn the patterns
2 FOUND.	8×100 @ CSS, 15s rest	8×50 drill · 8×100 @ CSS+6, hold count	12×25 fast · 6×100 @ CSS, tempo on	2.95s ~20 spm	A: +1 cord set · B: full prehab
3 BUILD	5×200 @ CSS, 20s rest	drills · 6×150 @ CSS+6	16×25 fast · 4×150 @ CSS	2.85s ~21 spm	A: power load up, cord 10×25s · B: +hinge load
4 BUILD	400-300-200-100 descend @ CSS → CSS-3, 20s	drills · 8×100 @ CSS+5, lowest count	8×50 @ CSS-5 · 200 @ CSS	2.85s ~21 spm	A & B progress; quality over load
5 SHARPEN	10×100 @ CSS, 12s rest, tempo on	drills · 6×150 @ CSS+4	16×25 @ CSS-5 · 4×100 @ CSS-2	2.75s ~22 spm	B-cord → 5×45s turnover-endurance · A power sharp
6 SHARPEN	6×200 @ CSS, 15s rest	drills · 8×100 @ CSS+4	12×50 @ CSS-5 · 200 @ CSS-2	2.75s ~22 spm	turnover-endurance emphasis (late-swim sag)
7 PEAK	8×100 @ CSS-2, 20s rest (quality)	short drills · 4×100 easy	8×25 sprint · 4×50 @ CSS-6	2.65s ~22.6 spm	cut volume; keep explosive power crisp
8 TAPER	easy · 4×100 @ CSS	easy technique swim	RE-TEST: CSS (400+200) + stroke count vs Week 1	free	one light session only

"@ CSS" scales to your Week-1 test · hold your stroke count on every tempo set · page 2 of 3

05 Session library

The standing structure each session hangs on. Pool sessions: ~300m warm-up, the main set from the table, ~150m easy cool-down. Dryland: ~30 min, twice weekly.

POOL · SESSION 1

Threshold / Tempo

Warm-up	300m mixed, build last 50
Activate	4×25 build to CSS, Tempo Trainer on from the week table, @ CSS on short rest
Main	hold stroke count; let the beep drive the hand
Cue	150m easy
Cool-down	

POOL · SESSION 2

Technique & aerobic

Warm-up	300m easy
Drills	6–8×50: single-arm, catch-up, fingertip-drag — count every length
Main	from table, easy-moderate, lowest smooth count
Goal	protect & extend distance/stroke; active recovery day
Cool-down	100m easy

POOL · SESSION 3

Speed & race-pace

Warm-up	300m + 4×25 build
Speed	short reps, full recovery — true top-end from table; race-pace reps with Tempo Trainer a notch up
Main	raise the ceiling so cruise pace feels easy
Goal	150m easy
Cool-down	

DRYLAND · A

Power & turnover

Prep	5 min: band dislocates, scap push-ups, open-books
Power	med-ball slams 3×8 + rotational throws 3×8/side, explosive
Turnover	cord stroke-rate intervals: 8×25s fast / 60s rest, metronome above pool cadence
Pull	pull-ups 4×6–8 (band ok), drive up fast
Core	Pallof press 3×10/side · hollow hold 3×30s

DRYLAND · B

Strength-endurance & resilience

Prep	5 min shoulder + thoracic mobility
Rate endur.	cord: 5×45s steady quick turnover / 75s rest (fixes late-swim sag)
Pull	bent-over row 3×10 · single-arm row 3×10
Posterior	RDL / KB hinge 3×10
Prehab	band ext. rotations 3×15 · face pulls 3×15 · Y-T-W 2×8
Core	side plank 3×30s/side · dead bug 3×8/side

GUARDRAILS

Make it stick, stay healthy

- **Speed over load** — light resistance moved fast builds turnover; heavy & slow doesn't.
- **Prehab is mandatory** — adding fast pulling volume is what triggers swimmer's shoulder. Sharp shoulder pain = stop.
- **Don't stack** the explosive dryland day before the speed swim; keep one easy day before the threshold swim.
- **One full rest day** a week — the gains land during recovery, not the session.
- **Land builds the engine; the pool spends it** — carry every cord turnover session into the water that week.